A Coordinate Anisotropic Lift of Schindl's No-Convex-Equivalent Weight

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Status

This is a partial result for the final question/goal in Schindl's paper [1]. The result gives an explicit anisotropic lift under the natural coordinate-log-convex analogue of the one-dimensional condition (ω_4). It does not claim to settle every possible convention for anisotropic Braun-Meise-Taylor weights.

Source Question

The source paper constructs in Section 7 a one-dimensional weight whose equivalence class contains no representative satisfying (ω_4), i.e. no equivalent σ for which $t \mapsto \sigma(e^t)$ is convex. It then asks for a higher-dimensional anisotropic analogue:

> with the conventions $0^0 := 1$ and $\log(0) = -\infty$. When treating this function, naturally one should assume that $\lim_{p\to+\infty}(M_p)^{1/p} = +\infty$ holds in order to ensure $\omega_{\mathbf{M}} : [0,+\infty) \to [0,+\infty)$ (well-definedness). For more details we refer to [16, Chapitre I], the more recent work [33] and see also [15, Sect. 3]. It is known and immediate by definition that (ω_4) holds for $\omega_{\mathbf{M}}$ and consequently in the whole equivalence class (w.r.t. relation $\sim$) of the previously constructed function there does not exist any associated weight function. In particular, the obtained (counter-)example cannot be described within the weight sequence setting which means that $\omega \equiv \omega_{\mathbf{M}}$ fails for any weight sequence $\mathbf{M}$.
> On the other hand the above constructed ω is matrix admissible since crucially (ω_3) holds.
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> We finish by mentioning the following *question/goal:* Transfer the construction of φ_ω (resp. of ω) in Section 7 to the higher dimensional case (anisotropic setting). We expect that the analogous statement holds as well but such an explicit construction is more technical and involved and also the arguments using the growth index $\gamma(\cdot)$ are becoming unclear in this general situation since so far this quantity is only defined and studied for $\omega : [0,+\infty) \to [0,+\infty)$.
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> References

The final paragraph asks to "transfer the construction of φ_ω (resp. of ω) in Section 7 to the higher dimensional case (anisotropic setting)" and notes that the growth-index argument is unclear in that setting.

Definitions and Source Input

For functions $F, G : \mathbb{R}^d \to [0, \infty)$ write $F \sim G$ if there is $C \geq 1$ such that

$$F(x) \leq CG(x) + C, \qquad G(x) \leq CF(x) + C \qquad (x \in \mathbb{R}^d).$$

For one-variable functions this is the equivalence relation used in [1].

For an anisotropic function $\Sigma : \mathbb{R}^d \to [0, \infty)$ define the coordinate-log-convex analogue of (ω_4) by requiring

$$\Phi_\Sigma(u_1, \dots, u_d) := \Sigma(e^{u_1}, \dots, e^{u_d})$$

to be convex on $\mathbb{R}^d$. This is the multivariable condition addressed here.

We use the output of Section 7 of [1]: there is a one-dimensional function ω with the following properties.

- ω is equivalent to a continuous, nondecreasing, subadditive function $\rho : [0, \infty) \to [0, \infty)$ with $\rho(0) = 0$.
- ρ satisfies (ω_3): for every $a \geq 1$ there is $b \in \mathbb{R}$ such that $\rho(t) \geq a \log(1 + t) + b$ for all $t \geq 0$.
- No one-dimensional function $\sigma \sim \rho$ has $t \mapsto \sigma(e^t)$ convex.
- The source construction gives uncountably many pairwise non-equivalent choices of such ρ.

The first and third bullets follow because Section 7 proves the obstruction for ω and also proves that ω is equivalent to the continuous subadditive representative κ_ω; equivalence preserves the obstruction.

Idea of the Proof

The lift is intentionally simple: put independent copies of the bad one-dimensional representative on the coordinate axes, but with unequal coefficients so the resulting weight is genuinely non-radial. Subadditivity and growth survive coordinatewise. If an equivalent anisotropic weight had the coordinate-log convexity property, freezing all coordinates except the first at 1 would produce a one-dimensional convex representative equivalent to ρ, exactly contradicting the Section 7 obstruction. Thus no new growth-index machinery is needed.

Theorem 1 (coordinate anisotropic lift)**.** *Let $d \geq 2$ and let ρ be as above. Define*

$$\Omega(x_1, \dots, x_d) := \rho(|x_1|) + 2\rho(|x_2|) + \sum_{j=3}^{d} \rho(|x_j|).$$

Then $\Omega : \mathbb{R}^d \to [0, \infty)$ is continuous, subadditive, and non-radial. It satisfies the anisotropic lower growth condition

$$\forall a \geq 1 \; \exists b \in \mathbb{R} \; \forall x \in \mathbb{R}^d : \qquad \Omega(x) \geq a \log(1 + |x|) + b.$$

Also, for every $\lambda > 0$,

$$\int_{\mathbb{R}^d} e^{-\lambda \Omega(x)} \, dx < \infty.$$

Finally, there is no $\Sigma : \mathbb{R}^d \to [0, \infty)$ such that $\Sigma \sim \Omega$ and $\Phi_\Sigma(u) = \Sigma(e^{u_1}, \dots, e^{u_d})$ is convex on $\mathbb{R}^d$.

Proof. Continuity and $\Omega(0) = 0$ are immediate from the corresponding properties of ρ. Since ρ is subadditive and nondecreasing,

$$\rho(|x_j + y_j|) \le \rho(|x_j| + |y_j|) \le \rho(|x_j|) + \rho(|y_j|),$$

and summing over the coordinates with the displayed positive coefficients gives $\Omega(x+y) \le \Omega(x) + \Omega(y)$.

The function is not radial. Indeed, for any t with $\rho(t) > 0$,

$$\Omega(t, 0, \ldots, 0) = \rho(t), \qquad \Omega(0, t, 0, \ldots, 0) = 2\rho(t),$$

although the two points have the same Euclidean norm.

For the lower growth, let $c_{\min} = 1$. Given $x \in \mathbb{R}^d$, choose a coordinate j with $|x_j| \ge |x|/\sqrt{d}$. Since every coefficient in Ω is at least 1,

$$\Omega(x) \ge \rho(|x_j|) \ge \rho(|x|/\sqrt{d}).$$

Given $a \ge 1$, use (ω_3) for ρ to choose b_0 such that $\rho(t) \ge a\log(1+t) + b_0$ for all $t \ge 0$. Since $1 + r/\sqrt{d} \ge (1+r)/\sqrt{d}$ for $r \ge 0$,

$$\Omega(x) \ge a\log(1+|x|) - \frac{a}{2}\log d + b_0.$$

This is the desired anisotropic (ω_3) condition.

Next fix $\lambda > 0$. The integral factors:

$$\int_{\mathbb{R}^d} e^{-\lambda\Omega(x)}\,dx = \prod_{j=1}^{d} \int_{\mathbb{R}} e^{-\lambda c_j \rho(|t|)}\,dt,$$

where $(c_1, c_2, c_3, \ldots, c_d) = (1, 2, 1, \ldots, 1)$. For each j, choose $q_j > 1/(\lambda c_j)$. By (ω_3), $\rho(t) \ge q_j \log(1+t) + b_j$. Hence

$$\int_{\mathbb{R}} e^{-\lambda c_j \rho(|t|)}\,dt \le 2e^{-\lambda c_j b_j} \int_0^\infty (1+t)^{-\lambda c_j q_j}\,dt < \infty.$$

Thus the product is finite.

It remains to prove the obstruction to coordinate-log convex representatives. Assume, toward a contradiction, that $\Sigma \sim \Omega$ and Φ_Σ is convex. Define a one-dimensional function

$$\sigma(t) := \Sigma(t, 1, \ldots, 1), \qquad t \ge 0.$$

Then

$$\Omega(t, 1, \ldots, 1) = \rho(t) + K,$$

where $K = 2\rho(1) + \sum_{j=3}^{d} \rho(1)$ is constant. The equivalence $\Sigma \sim \Omega$ therefore implies $\sigma \sim \rho$ as functions on $[0, \infty)$. But

$$s \mapsto \sigma(e^s) = \Sigma(e^s, 1, \ldots, 1) = \Phi_\Sigma(s, 0, \ldots, 0)$$

is the restriction of a convex function to a line, hence is convex. This contradicts the one-dimensional Section 7 obstruction for ρ. □

Theorem 2 (uncountably many lifts)**.** *The construction above gives uncountably many pairwise non-equivalent anisotropic weights Ω with the same properties.*

Proof. Take the uncountable family of pairwise non-equivalent one-dimensional representatives ρ_α supplied by Section 7 of [1], and form Ω_α as above. If $\Omega_\alpha \sim \Omega_\beta$, then restricting to $(t, 1, \dots, 1)$ gives

$$\rho_\alpha(t) + K_\alpha \sim \rho_\beta(t) + K_\beta,$$

and hence $\rho_\alpha \sim \rho_\beta$, contradicting the choice of the family. $\square$

Verification and Limitations

The proof is deterministic and uses no computation beyond rendering the source PDF and crop. The local run indexes and cheap source indexes were searched for `2510.05469`, `anisotropic convex`, `coordinate log convex`, `log-convex anisotropic`, and `Beurling Bjorck anisotropic`; no existing packet for this coordinate lift was found.

The result is intentionally scoped. It answers the final question under the coordinate-log-convex formulation stated above. If the intended anisotropic replacement for (ω_4) in another paper is weaker, stronger, or phrased through a different transform, that convention should be checked separately.

References

[1] G. Schindl. *On inclusion relations of weighted L^p-type spaces defined in terms of weight function matrices.* arXiv:2510.05469.